

Interoperable Web GIS for Air Quality Monitoring

GNR-629

GEOSPATIAL STANDARDS, INTEROPERABILITY AND KNOWLEDGE DISCOVERY

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PRESENTED TO

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— Introduction

- ① Geospatial interoperability allows different GIS systems to communicate using standard protocols.
- ② OGC standards such as WMS and WFS enable sharing of spatial data over the web.
- ③ This project demonstrates how a web client can interact with GeoServer to retrieve and display geospatial data.

Project Objective

**1 Develop an
interoperable
web-based GIS client.**

**2 Develop an
interoperable
web-based GIS client.**

**3 Retrieve spatial layers
using OGC standards
(WMS/WFS).**

**4 Visualize spatial layers
and service responses in
the web interface.**

**5 Demonstrate
interoperability with
external WMS
services.**

Application Domain: Urban Environmental Monitoring

①

Focus on visualizing geospatial information related to environmental data.

Example: Air quality monitoring in Mumbai

②

Spatial layers represent monitoring locations and interpolated raster surfaces.

③

GIS visualization helps understand spatial patterns in environmental conditions.

Technology Stack

- **Web Client:** HTML, CSS, JavaScript
- **Mapping Library:** OpenLayers
- **GIS Server:** GeoServer
- **Application Server:** Apache Tomcat
- **Spatial Database:** PostgreSQL + PostGIS
- **Data Processing:** QGIS

System Architecture

Client Layer:

- Web Browser
- OpenLayers + JavaScript web client

Service Layer:

- GeoServer running on Apache Tomcat
- Provides WMS and WFS services

Data Layer:

- PostgreSQL + PostGIS spatial database
- Stores vector and raster datasets

External Services:

- External WMS server (e.g., Bhuvan WMS)

Data Used in the Project

Vector Layers

e.g., monitoring stations,
boundaries

Raster Layers

e.g., interpolated
pollution surfaces

Data layers created in QGIS.

All layers converted to a consistent coordinate reference system.
Layers published through GeoServer and stored in PostGIS where required.

Web Client Interface (GUI)

- Interactive map interface built using OpenLayers.
- Base map using OpenStreetMap.
- Ability to add and display multiple layers.
- User options for selecting layers and sending service requests.
- Integration with local GeoServer services.

Web Client Interface (GUI)

GIS Web Client

WMS

WFS

Layers

WMS SERVER URL

http://localhost:8080/geoserver/wms

OK

LAYERS

Waiting for connection...

FORMAT

image/png

SRS

EPSG:3857

BOUNDING COORDINATES

Top / North

Left / West

Right / East

Bottom / South

FEATURE INFO

34

14

▶ RUN REQUEST

MAP VIEW

Lon: -171.3559, Lat: 42.6367

+
-
E



000 km

© OpenStreetMap contributors.

MAP DETAILS & LOGS

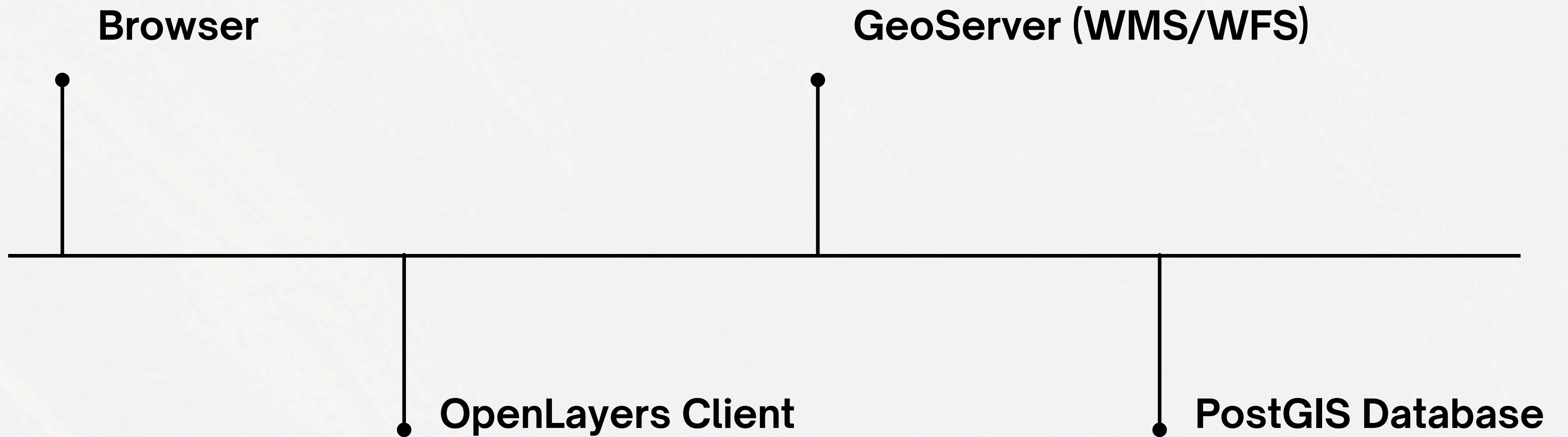
Table

Clear

Waiting server queries...

6

Workflow



Results

GIS Web Client

WMS

WFS

Layers

WMS SERVER URL

http://localhost:8080/geoserver/GIS_Proje

OK

LAYERS

PM2.5_MUMBAI (PM2.5_MUMBAI)

FORMAT

image/png

SRS

EPSG:4326

BOUNDING COORDINATES

19.2327

72.812972.9426

18.9099

FEATURE INFO

42313

RUN REQUEST

MAP VIEW

Lon: -169.3220, Lat: -27.6850

10 km

MAP DETAILS & LOGS

TableClear

[14:49:51] Successfully extracted 6 layers from document.

[14:49:58] Extracted Bounding Box for PM2.5_MUMBAI: [W:72.81285095214844, S:18.909921646118164, E:72.94256591796875, N:19.23268699645996]

[14:50:02] Executing GetMap for layer: PM2.5_MUMBAI

[14:50:02] Map view fitted to coordinates: [72.8129, 18.9099, 72.9426, 19.2327]

9

Results

GIS Web Client

WMS

WFS

Layers

WMS SERVER URL

http://localhost:8080/geoserver/GIS_Proj

OK

LAYERS

Mum_bound (Mum_bound)

FORMAT

image/png

SRS

EPSG:4326

BOUNDING COORDINATES

19.2696

72.7756

72.9811

18.8934

FEATURE INFO

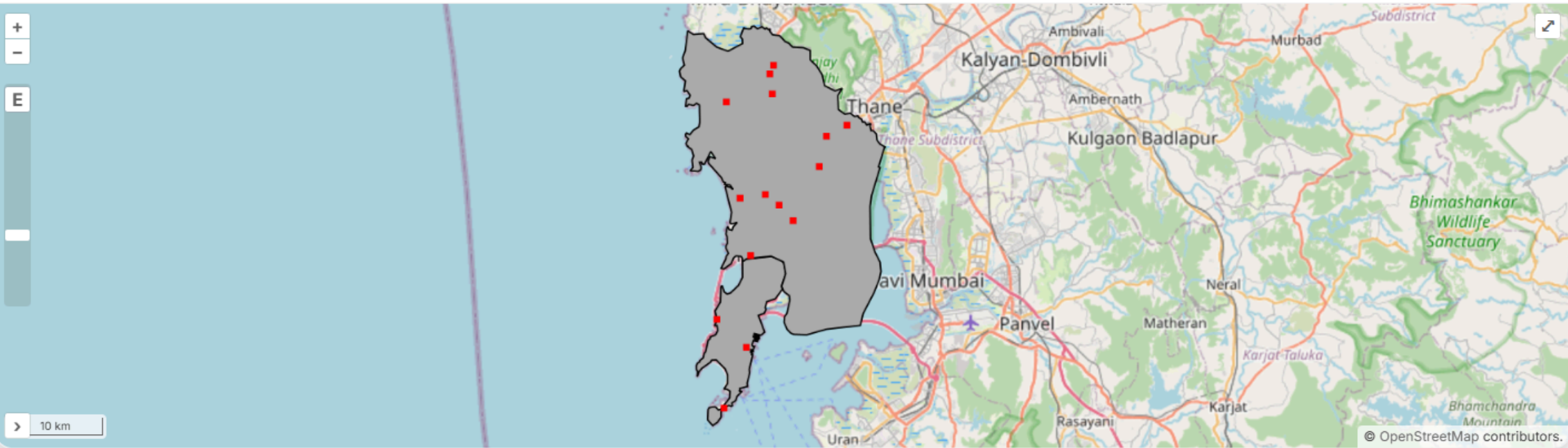
20

336

RUN REQUEST

MAP VIEW

Lon: 72.2232, Lat: 18.9487



MAP DETAILS & LOGS

Table

Clear

[14:50:02] Executing GetMap for layer: PM2.5_MUMBAI

[14:50:02] Map view fitted to coordinates: [72.8129, 18.9099, 72.9426, 19.2327]

[14:51:27] Extracted Bounding Box for Mum_bound: [W:72.77557373046875, S:18.893404006958008, E:72.98113250732422, N:19.269611358642578]

[14:51:28] Executing GetMap for layer: Mum_bound

[14:51:29] Map view fitted to coordinates: [72.7756, 18.8934, 72.9811, 19.2696]

9

Results

GIS Web Client

WMS

WFS

Layers

WMS SERVER URL

http://localhost:8080/geoserver/GIS_Proje

OK

LAYERS

interpolated_PM2.5_final (interpolated_PM2.5_fin

FORMAT

image/png

SRS

EPSG:4326

BOUNDING COORDINATES

19.2696

72.7756

72.9811

18.8934

FEATURE INFO

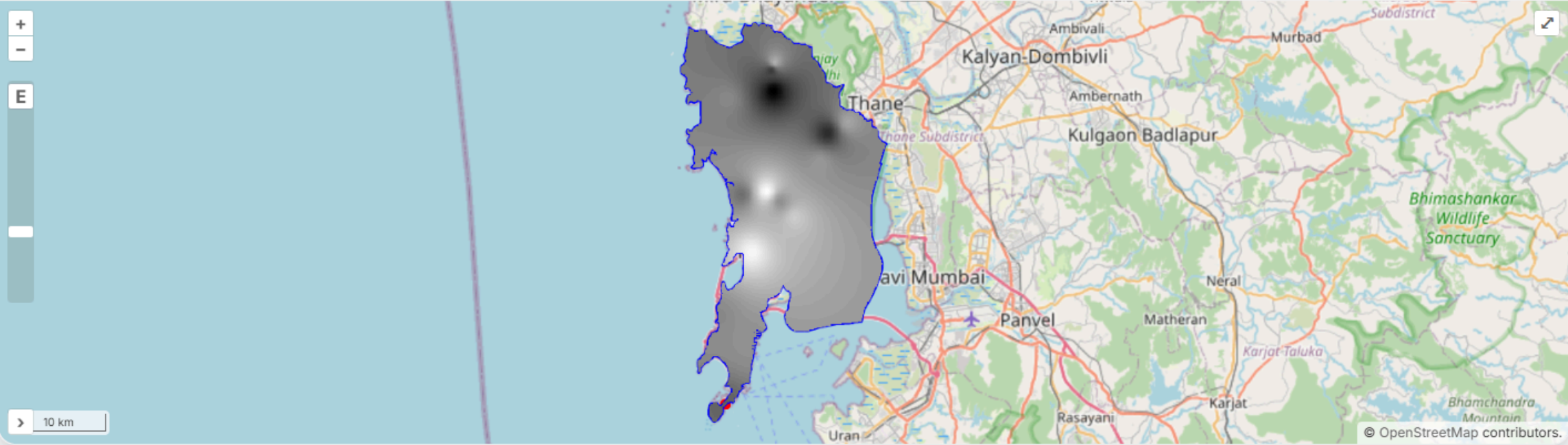
711

191

RUN REQUEST

MAP VIEW

Lon: 72.8817, Lat: 19.0907



MAP DETAILS & LOGS

Table

Clear

```
"timeStamp": "2026-03-09T09:22:50.800Z",
"crs": {
  "type": "name",
  "properties": {
    "name": "urn:ogc:def:crs:EPSG::3857"
  }
}
}
```

[14:52:50] Map view fitted to GeoJSON feature extent.

[14:52:57] Fetching Feature Info for Map Click at pixel [734, 135]...

9

Results

GIS Web Client

WMS

WFS

Layers

WFS SERVER URL

http://localhost:8080/geoserver/GIS_Proj

OK

FEATURE TYPES (LAYERS)

PM2.5_MUMBAI (GIS_Project_1:PM2.5_MUMBAI) ▾

FORMAT

GeoJSON ▾

SRS

EPSG:3857 ▾

BOUNDING COORDINATES

19.2327

72.812972.9426

18.9099

▶ RUN WFS REQUEST

MAP VIEW

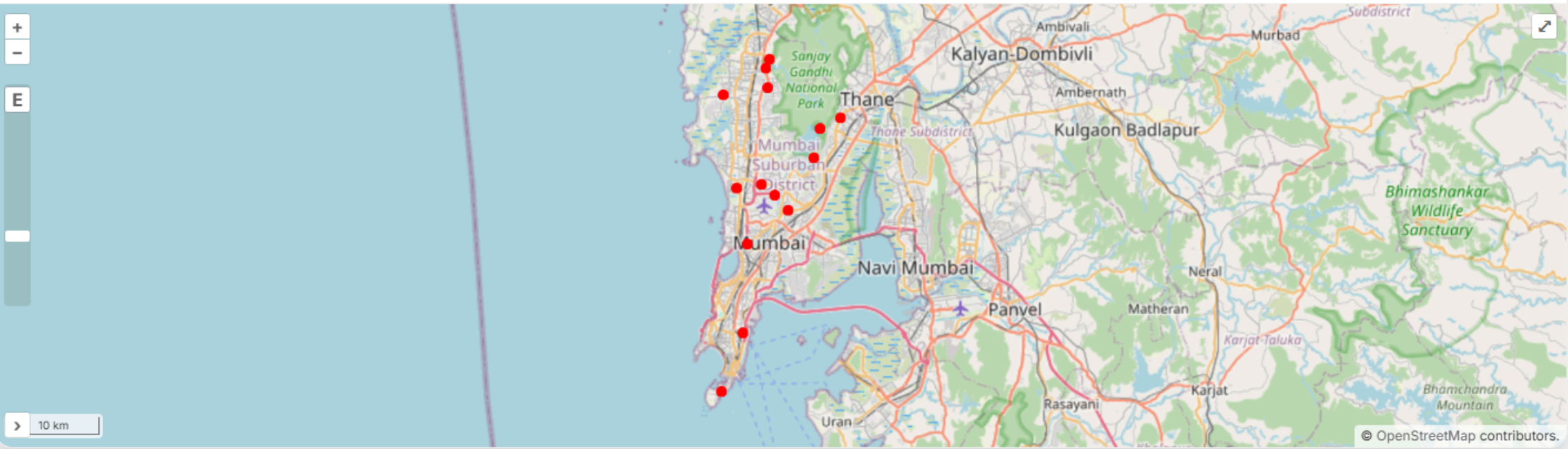
Lon: 72.0738, Lat: 19.1790

+

-

E

10 km



MAP DETAILS & LOGS

Logs

Clear

Attribute	Value
station_name	Powai
pm_2.5	110
latiitude	19.13742233
longitude	72.91499706

Results

GIS Web Client

WMS

WFS

Layers

WFS SERVER URL

http://localhost:8080/geoserver/GIS_Proje

OK

FEATURE TYPES (LAYERS)

Mum_bound (GIS_Project_1:Mum_bound)

FORMAT

GeoJSON

SRS

EPSG:3857

BOUNDING COORDINATES

19.2696

72.7756

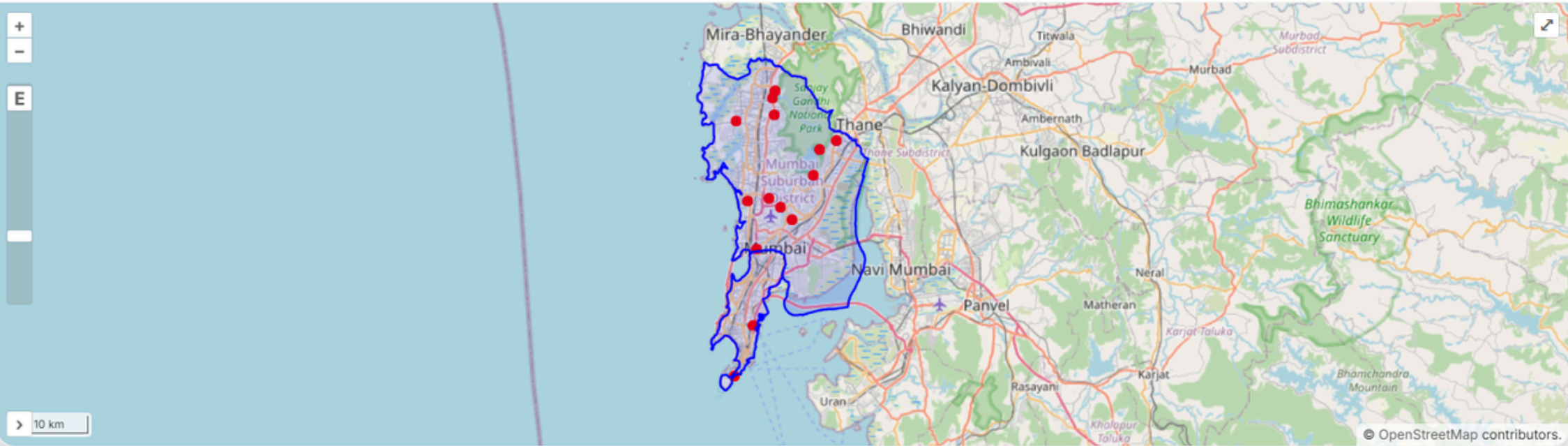
72.9811

18.8934

▶ RUN WFS REQUEST

MAP VIEW

Lon: 72.9542, Lat: 19.1064



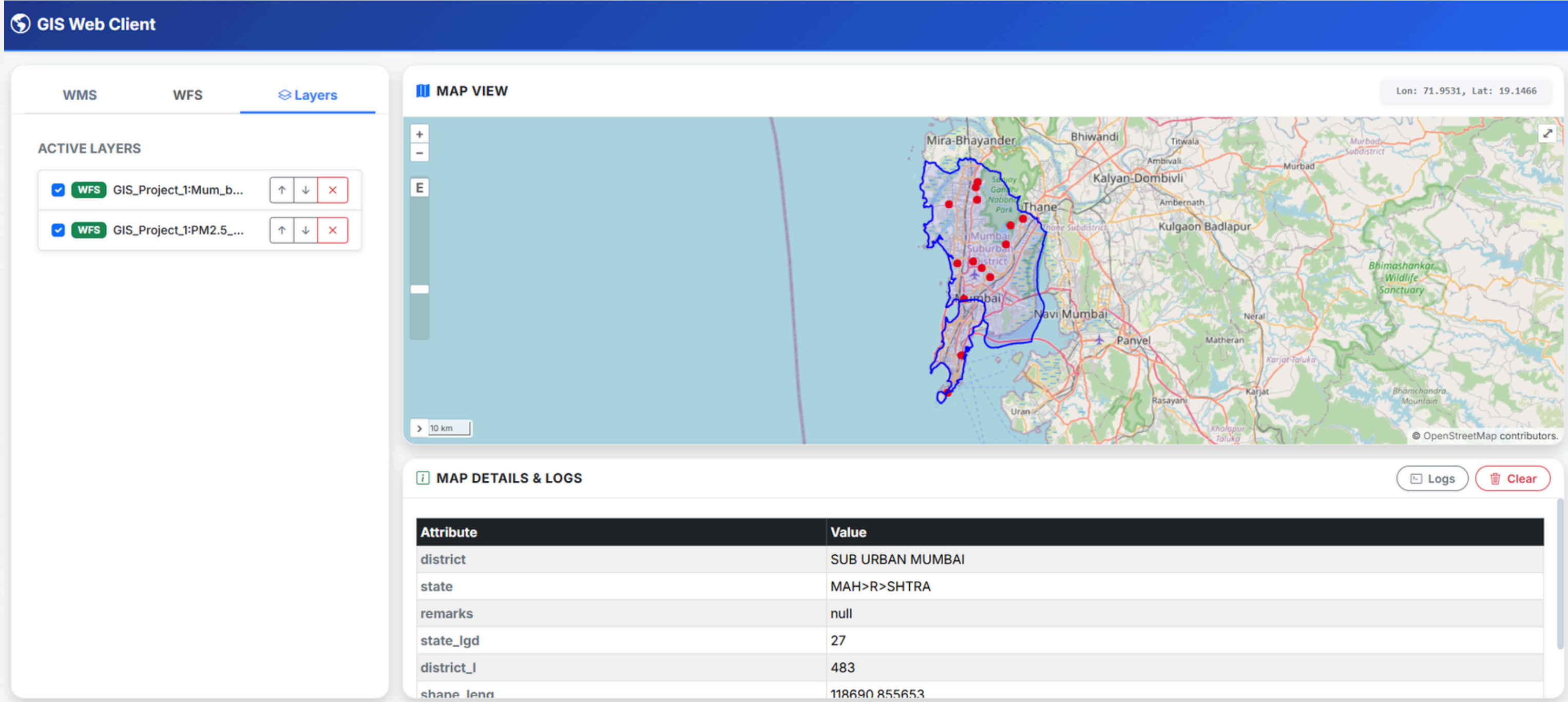
MAP DETAILS & LOGS

Logs

Clear

Attribute	Value
district	SUB URBAN MUMBAI
state	MAH>R>SHTRA
remarks	null
state_lgd	27
district_l	483
shape_leng	118690.855653

Results



Conclusion

- Developed a client–server interoperable GIS application.
- Implemented OGC standards (WMS/WFS) for geospatial data access.
- Integrated GeoServer, PostGIS, and OpenLayers.
- Demonstrated how web technologies enable sharing and visualization of geospatial data.

Project 2

Objective

Develop an SOS-based system

Handle real-world-like sensor data

Implement:

- SensorML
- Spatial filtering
- Temporal filtering
- Attribute filtering

Build a web client for visualization

Technology Stack

Frontend:

- HTML5
- CSS (Bootstrap)
- JavaScript
- OpenLayers

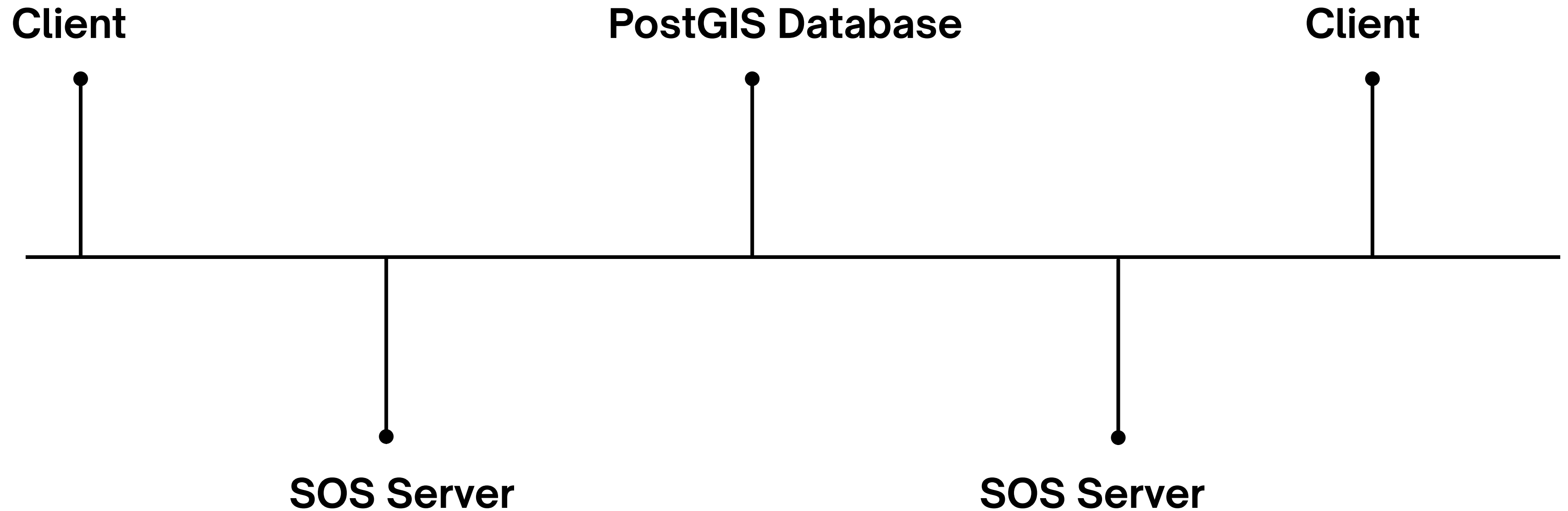
Backend:

- istSOS (SOS server)
- Python (Waitress server)

Database:

- PostgreSQL

Workflow



SOS Interface (GUI)

GIS Web Client

WMS

WFS

SOS

Layers

SOS SERVER URL

http://localhost:8090/istsos/mumbai

OK

REQUEST TYPE

GetObservation

PROCEDURES / SENSORS

Waiting for connection...

OBSERVED PROPERTY

PM2.5

SPATIAL BOUNDING BOX

Top / North

Left / West

Right / East

Bottom / South

TEMPORAL SUBSETTING

dd-mm-yyyy

dd-mm-yyyy

VALUE FILTER

No Filter

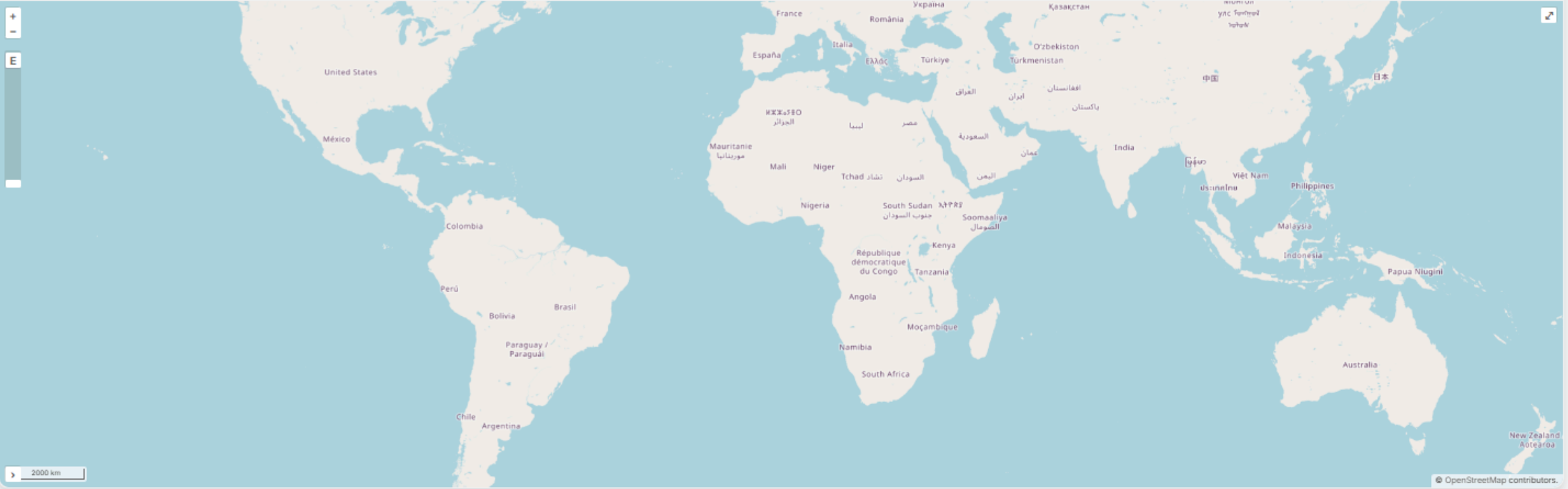
Value

▶ RUN SOS REQUEST

VIEW SENSORML

MAP VIEW

Lon: -156.1361, Lat: 32.0649

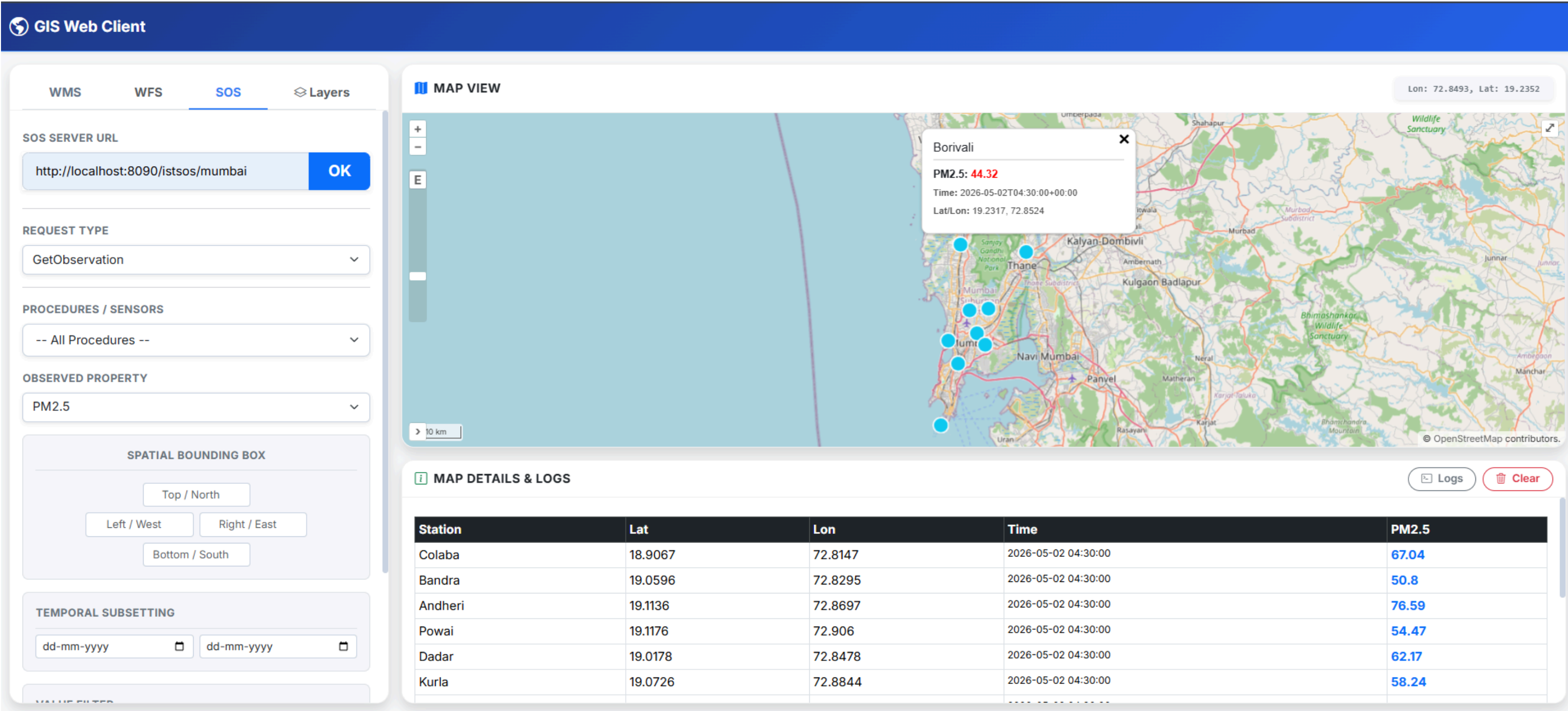


MAP DETAILS & LOGS

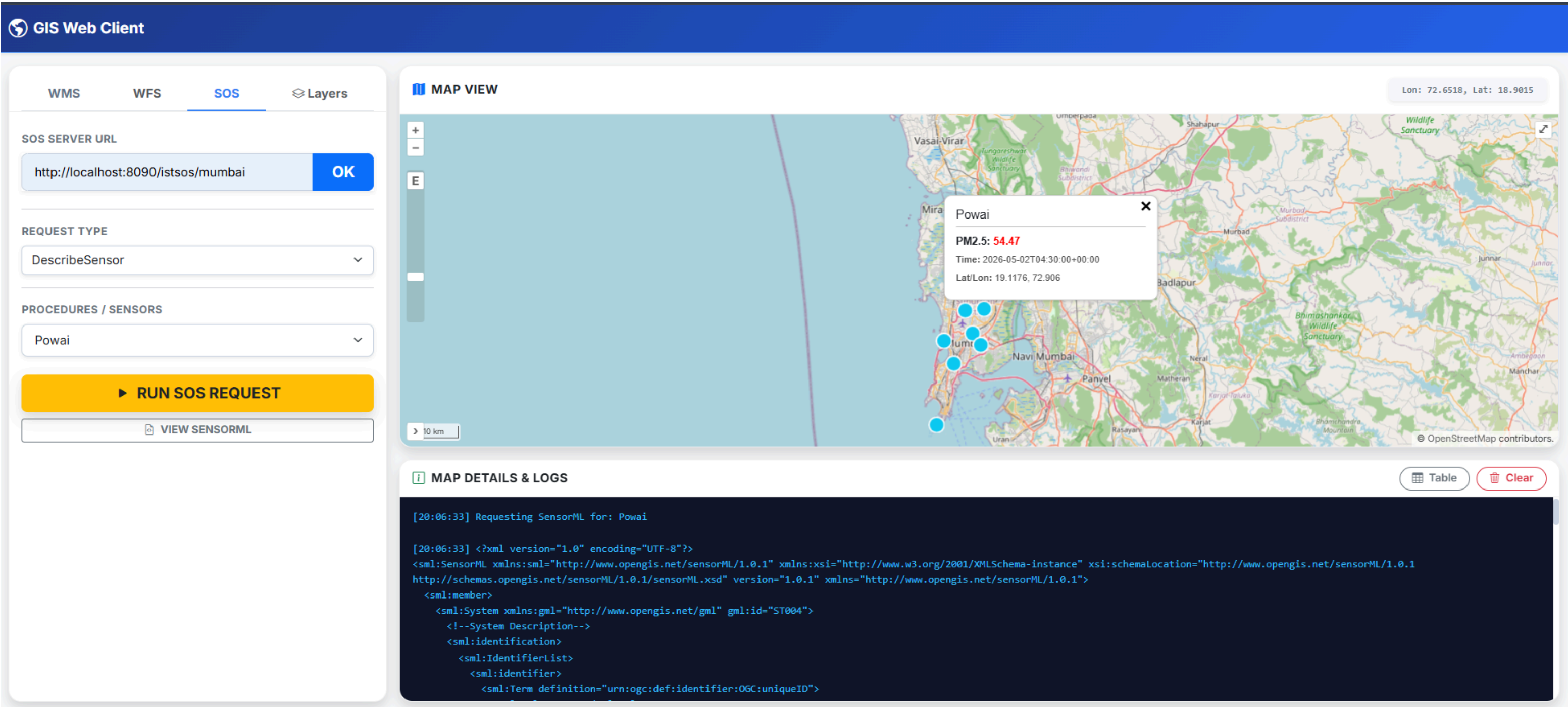
Table

Clear

Results



Results



Results

GIS Web Client

WMS

WFS

SOS

Layers

SOS SERVER URL

http://localhost:8090/istsos/mumbai

OK

REQUEST TYPE

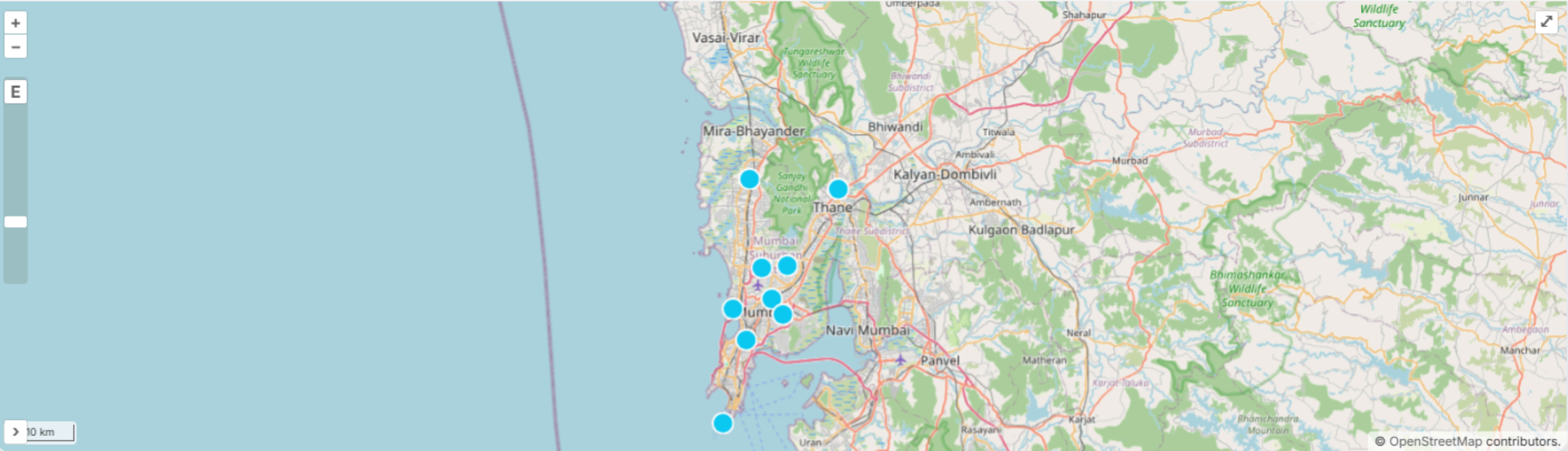
GetCapabilities

▶ RUN SOS REQUEST

VIEW SENSORML

MAP VIEW

Lon: 73.1189, Lat: 19.3253



10 km

MAP DETAILS & LOGS

Table

Clear

[20:08:18] Sending SOS GetCapabilities request to:
http://localhost:8090/istsos/mumbai

[20:08:18] Received SOS GetCapabilities response. Parsing XML...

[20:08:18] <?xml version="1.0" encoding="UTF-8"?>
<Capabilities
 xmlns:gml="http://www.opengis.net/gml"
 xmlns:xlink="http://www.w3.org/1999/xlink"
 xmlns:swe="http://www.opengis.net/swe/1.0.1"
 xmlns:om="http://www.opengis.net/om/1.0"
 xmlns="http://www.opengis.net/sos/1.0"
 xmlns:sos="http://www.opengis.net/sos/1.0"

The End

THANK YOU FOR LISTENING